CSM GLEAMING Solutions

COMPUTER SCIENCE MENTORS 61A

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1 Gleam basics

Gleam is a functional programming language. Functional programming is a paradigm where programs are constructed by applying and composing functions, and function definitions are trees of expressions that map values to other values, rather than a sequence of statements which update the running state of the program. Gleam is a statically-typed, type-safe language. Type safety is the extent to which a programming language discourages or prevents type errors. The behaviors classified as type errors by a given programming language are usually those that result from attempts to perform operations on values that are not of the appropriate data type, e.g. trying to add a string to an integer.

Type enforcement can be static (catching potential errors at compile time), dynamic (associating type information with values at run-time and consulting them as needed to detect imminent errors), or a combination of both. In the context of static type systems, type safety usually involves a guarantee that the eventual value of any expression will be a legitimate member of that expression's static type.

When writing functions that operate on a given input, Gleam requires that you handle any theoretically possible output case, even if you've potentially handled such a case earlier. For instance, the function `list.max` (which behaves very similarly to Python's `max` function) does not return an integer directly, but a `Result`. This handles the case in which an empty list has been passed into `max`: it returns `Ok(value)` or `Error(Nil)` regardless of whether you've already handled the empty case earlier in your code, and are thus guaranteeing that it will always be an `Ok` result.

When writing a `case` statement in Gleam, you must handle every theoretically possible case, even if that case may not be possible in the context of your own code. For instance, say you have a function that exclusively outputs even integers, and you use one of those integers `n` in a `case` statement:

```
case n % 2 {
    0 -> //some code for the case in which n is even
    1 -> //some code for the case in which n is odd
}
```

Gleam will require you to include the latter case, in which `n % 2 == 1`, regardless of whether your integer `n` could possibly be odd.

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1. Implement `multiply` without using the `*` operator.

```
pub fn multiply(n: Int, m: Int) -> Int {
  case n > 0 {
    True -> _______ + _____________________________
    False -> _______
  }
}
```

```
pub fn multiply(n: Int, m: Int) -> Int {
  case n > 0 {
    True -> m + multiply(n - 1, m)
    False -> 0
  }
}
```

2. Implement `sum`, a tail recursive function that returns the sum of all elements of a list of integers.

```
fn sum_helper(l: List(Int), curr: Int) -> Int {
  case l {
    [] -> _______
    [first] -> _______ + _______
    [first, ..rest] -> _____________________________
  }
}

pub fn sum(l: List(Int)) -> Int {
  sum_helper(_______, _______)
}
```

```
fn sum_helper(l: List(Int), curr: Int) -> Int {
  case l {
    [] -> curr
    [first] -> curr + first
    [first, ..rest] -> sum_helper(rest, curr + first)
  }
}

pub fn sum(l: List(Int)) -> Int {
  sum_helper(l, 0)
}
```

3. Implement `reduce`, a tail recursive function that applies a two-argument function `f` on the elements of a list from left to right, starting with `start`.

```
fn reduce(l: List(a), f: fn(a, a) -> a, start: a) -> a {
  case l {
    [] -> _______
    [first] -> ___________________________________
    [first, ..rest] -> ___________________________________
  }
}
```

```
fn reduce(l: List(a), f: fn(a, a) -> a, start: a) -> a {
  case l {
    [] -> start
    [first] -> f(start, first)
    [first, ..rest] -> reduce(rest, f, f(start, first))
  }
}
```

```
pub type Tree(a) {
    Tree(label: a,
        branches: List(Tree(a)))
}

pub fn leaf(label: a) -> Tree(a) {
    Tree(label, [])
}
```

4. Implement `prune_even`, which prunes every branch in a tree that has an even label. Odd child branches of branches with even labels are not preserved.

```
pub fn prune_even(tree: Tree(Int)) -> Tree(Int) {
    let Tree(label, branches) = tree

    let remove_even = list.filter(branches, _____________________________)
    let prune_children = list._________(___________________________________)

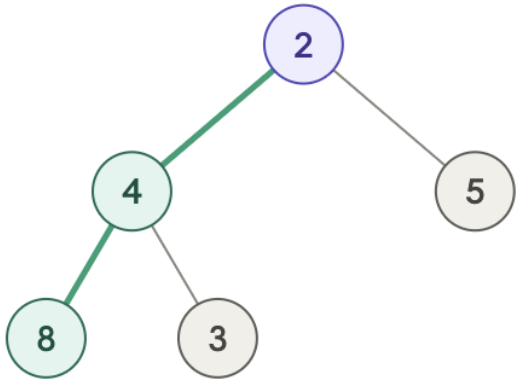
    Tree(label, prune_children)
}
```

```
pub fn prune_even(tree: Tree(Int)) -> Tree(Int) {
    let Tree(label, branches) = tree

    let remove_even =
        list.filter(branches, fn(branch) { branch.label % 2 != 0 })
    let prune_children = list.map(remove_even, prune_even)

    Tree(label, prune_children)
}
```

5. A *treemil* is a tree in which at least one of the child node(s) of each non-leaf node has a label that is a multiple of the label of the node. Alternatively, a treemil is a tree where every node has at least one path from the node to a leaf consisting entirely of nodes that are multiples of its parent. Implement `is_treemil`, which returns whether a tree is a treemil.




```
pub fn is_treemil(tree: Tree(Int)) -> Bool {
    // valid if:
    // 1. at least one child node is a multiple of its parent
    // 2. all children are valid treemils
    // leaves are valid treemils

    case tree {
        //is a leaf
        Tree(_, []) -> _________

        //is not a leaf
        Tree(label, branches) -> {
            let child_multiples = list._______(_______, ________________)
            case list._________(child_multiples) >= 1 {
                True -> list._________(___________________________________)
                False -> _________
            }
        }
    }
}
```

```
pub fn is_treemil(tree: Tree(Int)) -> Bool {
    case tree {
        Tree(_, []) -> True

        Tree(label, branches) -> {
            let child_multiples =
                list.filter(branches, fn(b) { b.label % label == 0 })
            case list.length(child_multiples) >= 1 {
                True -> list.all(list.map(branches, is_treemil), fn(x) { x
                    })
                False -> False
            }
        }
    }
}
```

6. Implement `max_treemil` which takes in a tree `t` and determines the highest number of nodes in a treemil that is a subtree of `t`. You may assume `is_treemil` is implemented correctly from the previous question.

```
pub fn max_treemil(tree: Tree(Int)) -> Int {
    //Given a tree (that is not necessarily a treemil), returns
    //the highest number of nodes in a treemil in the tree.

    case tree {
        //is a leaf
        Tree(_, []) -> _________

        //is not a leaf
        Tree(_, branches) -> {
            case ___________________________________ {
                True -> 1 + int._________(___________________________________)
                False -> result.unwrap(list.max(_________, int.compare), 0)
            }
        }
    }
}
```

```
pub fn max_treemil(tree: Tree(Int)) -> Int {
    case tree {
        Tree(_, []) -> 1

        Tree(_, branches) -> {
            case is_treemil(tree) {
                True -> 1 + int.sum(list.map(branches, max_treemil))
                False -> {
                    let sizes = list.map(branches, max_treemil)
                    result.unwrap(list.max(sizes, int.compare), 0)
                }
            }
        }
    }
}
```